

Code-Layout, Readability and Reusability

Date	26 June 2026
Team ID	SWIID-2026-1243
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

EduGenie follows a modular software architecture to ensure clean, maintainable, and reusable code. The project is organized into separate frontend and backend modules, with FastAPI handling backend APIs and HTML, CSS, and JavaScript providing the user interface. AI functionalities such as question answering, concept explanation, quiz generation, summarization, and learning recommendations are implemented as reusable services. Proper naming conventions, comments, exception handling, and modular design make the application scalable and easy to maintain.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained throughout the project.	Yes	Follows Python PEP 8 and consistent HTML/CSS formatting.
2	Proper File Structure	Files and folders are organized into frontend, backend, AI services, and utilities.	Yes	Modular folder structure improves maintainability.
3	Meaningful Variable Names	Variables use descriptive and meaningful names.	Yes	Improves readability and debugging.
4	Function / Method Names	Functions are named according to their specific responsibilities.	Yes	Easy to understand and reuse.
5	Code Comments		Yes	Enhances code understanding for team members.

		Important functions and logic include comments and documentation.		
6	Modular Design	Code is divided into reusable modules and API services.	Yes	Separate modules for AI, APIs, frontend, and utilities.
7	No Redundant Code	Duplicate code is minimized using reusable functions.	Yes	Common logic is shared across modules.
8	Error Handling	Exceptions and API errors are handled gracefully.	Yes	Validation and exception handling implemented in FastAPI.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	FastAPI API Router	Python, FastAPI	All API endpoints	High
2	Gemini AI Service	Python, Google Gemini 1.5 Pro	Question Answering, Explanation, Quiz, Summary, Roadmap	High
3	LaMini-Flan-T5 Service	Python, Transformers	AI response generation	High
4	Prompt Engineering Module	Python	All AI features	High
5	Frontend Components	HTML, CSS, JavaScript	User Interface	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
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Code Layout & Structure	5	Well-organized frontend, backend, and AI modules.
Readability	5	Meaningful naming conventions and consistent formatting.
Reusability	5	Modular services and reusable API components.
Documentation / Comments	5	Code includes comments and project documentation.
Overall Score	5 / 5	The project follows clean coding practices, modular architecture, and reusable design principles